Sharp gradient bounds and step selection under relative errors

Abstract

We study constant-step gradient descent when each computed gradient has a deterministic relative error below one. An explicit convex function of two variables attains the published intermediate one-step bound, completing its exact worst-case curve and determining its unique optimal step. The error in an optimal attaining instance is perpendicular to the computed direction. For multiple updates, we determine the exact asymptotic worst-case minimum squared gradient norm under an initial function-gap condition. A local interpolation certificate with a telescoping gradient correction gives the upper bound; explicit trajectories attain its leading coefficient. We obtain the exact asymptotic minimax step and convergence of asymptotically minimizing step sequences. For the last gradient, a staged family in logarithmically growing dimensions rules out a dimension-independent $O(1/N)$ bound at every fixed normalized step $0 < h \leq 1$. In the plane, we prove such a bound for $0 < h \leq 1/(1+\delta)$, including its endpoint. The proof controls gradient-magnitude records along the actual polygonal path and also gives a continuous-time estimate.

Keywords: inexact gradient descent; relative error; sharp bounds; step selection; smooth convex interpolation

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1 The performance problem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, with an L-Lipschitz gradient, where $L > 0$ and d is any positive integer. Assume that f attains its minimum f_*. Fix $0 < \delta < 1$ and a normalized step $h \geq 0$. We consider

$$x_{k+1} = x_k - \frac{h}{L} d_k, \qquad \|d_k - \nabla f(x_k)\| \leq \delta \|\nabla f(x_k)\|. \tag{1}$$

The directions are arbitrary deterministic vectors satisfying the displayed condition at each update. The same h is used throughout a run.

For $N \geq 1$, define

$$\mathcal{W}_N(h, \delta) = \sup \frac{\min_{1 \leq k \leq N} \|\nabla f(x_k)\|^2}{L(f(x_0) - f_*)}, \tag{2}$$

where the supremum is over all the preceding data with $f(x_0) > f_*$. Thus the dimension, objective, initial point and errors may depend on N. The initial point is excluded from the minimum, and the final iterate is included. For $N = 1$, this is the terminal-gradient criterion.

For this error model, Vernimmen and Glineur [1] derive an upper bound with three one-step regimes. Their scalar examples attain the two outer regimes; Conjecture 2.6 asks for tightness and bivariate attainment in the intermediate regime. We use the theorem and conjecture numbering of the public version listed in the bibliography. The accepted manuscript dated 31 October 2025 retains that conjecture. The interpolation inequalities and the positive-square structure behind the upper bound are prior work; our first result supplies a global attaining function throughout the intermediate interval and its exact step-selection geometry.

The same source gives a multiple-step minimum-gradient upper bound and numerical evidence concerning useful constant steps. Our second result identifies the exact asymptotic coefficient for the specific criterion (2), and then optimizes that coefficient. It does not give an exact finite-N curve or a multiple-step terminal-gradient upper bound. The initial condition is a function gap, not a distance to a minimizer.

For the last gradient, Vernimmen's thesis [4, Section 2.2] already studies one-step amplification, gives a non-tight last-gradient upper estimate, and compares the last and best gradients numerically. We retain that ancestry. Section 5 proves a dimension-independent rate obstruction using a prescribed error schedule on globally defined Huber objectives. The dimension and initial point vary with the horizon; the result makes no fixed-dimension or fixed-orbit assertion. Section 6 proves a class-uniform final-gradient bound in dimension two at the same smaller steps $0 < h \le 1/(1+\delta)$. The normalized unit step in the plane remains unresolved.

Planar geometry has an established role in descent: Daniilidis, Deville and Durand-Cartagena [5, Theorem 6.4] prove rectifiability of bounded planar curves satisfying a cone condition. Exact-gradient path-length bounds in [6] depend on spatial distances. Our planar argument instead controls the running gradient magnitude using the order in which gradient rays are visited. It requires no bound on the spatial trajectory.

Smooth convex interpolation [2] provides the basic inequalities used below. Earlier robustness results for strongly convex objectives, including the exponential estimates in [3], concern a different performance problem. The arguments here require neither strong convexity nor a bounded sublevel set.

2 The exact one-step curve

Put $D = \delta^2$ and define

$$q_R = \frac{2+\delta-\sqrt{4-3\delta^2}}{2}, \quad h_L = \frac{3}{2(1+\delta)}, \quad h_R = \frac{3\delta+2-\sqrt{4-3\delta^2}}{2\delta(1+\delta)}. \tag{3}$$

For $q_R \le q \le \delta$, write

$$V = 2q - q^2 - D, \quad U = 3q - q^2 - 2D, \quad h(q) = \frac{(1-q)U}{(1-D)V}, \quad \rho(q) = \frac{(1-q)U}{2q(1-D)}. \tag{4}$$

Theorem 1. *The map $q \mapsto h(q)$ is a continuous strictly decreasing bijection from $[q_R, \delta]$ onto $[h_L, h_R]$. For $0 \le h \le 2/(1+\delta)$,*

$$\mathcal{W}_1(h,\delta) = \begin{cases} [h(1-\delta)+\frac{1}{2}]^{-1}, & 0 \le h \le h_L, \\ [\rho(q)+\frac{1}{2}]^{-1}, \quad h = h(q), & h_L \le h \le h_R, \\ 2[1-h(1+\delta)]^2, & h_R \le h \le 2/(1+\delta). \end{cases} \tag{5}$$

The endpoint values agree. Every value is attained by a globally smooth convex function in dimension at most two.

The outer regimes and the intermediate upper bound are those of [1, Theorems 2.3–2.4]. The exact attainment below establishes its Conjecture 2.6. We give the upper certificate as well, so that the equality is verified directly.

2.1 Interpolation certificates

Dividing f and the directions by L, and subtracting f_*, reduces the analysis to $L = 1$ and $f_* = 0$. At one step write $g = \nabla f(x)$, $g_+ = \nabla f(x_+)$ and $e = d - g$, where $x_+ = x - hd$. The residuals

$$R_{01} = f(x) - f(x_+) - h\langle g_+, d\rangle - \tfrac{1}{2}\|g - g_+\|^2, \tag{6}$$

$$R_{10} = f(x_+) - f(x) + h\langle g, d\rangle - \tfrac{1}{2}\|g - g_+\|^2, \tag{7}$$

$$I = D\|g\|^2 - \|e\|^2 \tag{8}$$

are nonnegative. The first two are smooth convex interpolation inequalities; the third is the error condition. Also

$$f(x) - f_* \ge \tfrac{1}{2}\|\nabla f(x)\|^2. \tag{9}$$

For completeness, the interpolation inequality follows by subtracting the supporting affine function at one point and applying the smooth descent inequality to the resulting nonnegative convex function at the other point.

For the intermediate parameters (4), set

$$\lambda = \frac{q - D}{q(1-q)}, \qquad \beta = \frac{h}{2q}. \tag{10}$$

Direct expansion gives

$$\begin{aligned} f(x) - f(x_+) - \rho\|g_+\|^2 = (\lambda+1)R_{01} + \lambda R_{10} + \beta I \\ + \beta\|e + q((\lambda+1)g_+ - \lambda g)\|^2. \end{aligned} \tag{11}$$

The coefficient calculation uses precisely

$$D = q[1 - \lambda(1-q)], \quad h(\lambda+1)(1+q\lambda) = 2\lambda + 1, \quad \rho = \tfrac{h}{2}(\lambda+1)(1-q). \tag{12}$$

Once the parameter signs are checked below, (11) and (9) prove the intermediate upper bound.

In the left regime take $\lambda = 1$, $q = \delta$ and $\beta = h/(2\delta)$. The exact identity is

$$\begin{aligned} f(x) - f(x_+) - h(1-\delta)\|g_+\|^2 = 2R_{01} + R_{10} + \frac{h}{2\delta} I \\ + \frac{h}{2\delta}\|e + \delta(2g_+ - g)\|^2 \\ + \left(\frac{3}{2} - h(1+\delta)\right)\|g - g_+\|^2. \end{aligned} \tag{13}$$

Every coefficient is nonnegative for $0 \le h \le h_L$, including $h = 0$.

For the right regime let $r = h(1+\delta) - 1$. Then

$$r \in [q_R/\delta, 1], \quad \lambda = \frac{1-r}{r}, \quad \beta = \frac{h}{2\delta r}, \quad \rho = \frac{r^{-2} - 1}{2}.$$

One has

$$\begin{aligned} f(x) - f(x_+) - \rho\|g_+\|^2 = (\lambda+1)R_{01} + \lambda R_{10} + \beta I \\ + \beta\|e + \delta r((\lambda+1)g_+ - \lambda g)\|^2 + \xi\|g_+ + rg\|^2, \\ \xi = \frac{-\delta r^2 + (2+\delta)r - (1+\delta)}{2r^2(1+\delta)}. \end{aligned} \tag{14}$$

The numerator of ξ vanishes at $r = q_R/\delta$ and has positive derivative $2 + \delta - 2\delta r$ on the stated interval. Thus this certificate is nonnegative, including the upper endpoint. Adding (9) proves the remaining upper bound in (5).

2.2 Parameter signs and the global extremizer

Consider $T(q) = 2q - q^2 - D - \delta(1-q)$. It is strictly increasing on $[D, \delta]$, is negative at D, and is positive at δ. Its zero is q_R, so

$$D < q_R < \delta < 1, \qquad V \ge \delta(1-q) > 0, \qquad U = V + q - D > 0. \tag{15}$$

Also $T(\delta/2) = -3D/4 < 0$, hence $q_R > \delta/2$. Differentiation gives

$$h'(q) = -\frac{V^2 - (1-D)(D-q^2)}{(1-D)V^2} < 0, \tag{16}$$

because the numerator is at least $D(1-q)^2 - (1-D)(D-q^2) = (q-D)^2 > 0$. The endpoints are $h(\delta) = h_L$ and $h(q_R) = (1+q_R/\delta)/(1+\delta) = h_R$. This proves the bijection and $0 < h_L < h_R < 2/(1+\delta)$. In particular, the coefficients in (11) are positive.

Define

$$A = \frac{D + q^2 - 3q + 1}{1-D}, \qquad B = \frac{D-q}{V}, \qquad C = \frac{(q-D)(2-q)}{1-D}, \tag{17}$$

and

$$a = \sqrt{C^2 - DA^2}, \qquad z = \sqrt{D - q^2}. \tag{18}$$

Both square roots are real, since

$$C^2 - DA^2 = \frac{V^2 - D(1-q)^2}{1-D} \geq 0. \tag{19}$$

Here $a = 0$ only at $q = q_R$, and $z = 0$ only at $q = \delta$. They therefore never vanish simultaneously.

For $a \geq 0$, let

$$\phi_a(u) = \begin{cases} u^2/2, & |u| \leq a, \\ a|u| - a^2/2, & |u| \geq a, \end{cases} \qquad \phi_0 \equiv 0. \tag{20}$$

The function

$$f(u, v) = \phi_a(u) + \tfrac{1}{2}v^2 \tag{21}$$

is globally convex and continuously differentiable. Its gradient is $(\mathrm{clip}_{[-a,a]}(u), v)$, hence is 1-Lipschitz. Its minimum is zero and is attained at the origin.

Take

$$x_0 = ((1 + h(1-q))a, Az), \qquad d_0 = (1-q)(a, z). \tag{22}$$

The identities

$$A - B = h(1-q), \quad C = (1-q) - A, \quad -\lambda A + (\lambda + 1)B = -C/q \tag{23}$$

give

$$x_1 = (a, Bz), \quad g_0 = (a, Az), \quad g_1 = (a, Bz), \quad e_0 = (-qa, Cz).$$

The first coordinate of x_0 is at least a. Moreover

$$\|e_0\|^2 - D\|g_0\|^2 = (q^2 - D)a^2 + (C^2 - DA^2)z^2 = 0.$$

Both interpolation residuals vanish, because the first coordinate travels in an affine part of ϕ_a and the second has unit quadratic curvature. The square in (11) vanishes by (23), and $f(x_1) = \|g_1\|^2/2$. Thus the bound is attained. The construction remains valid if $A = 0$, since no division by A is used. In the open intermediate interval, $a, z > 0$ and $A - B > 0$, so g_0, g_1 are linearly independent.

For the left branch, use ϕ_1, $x_0 = 1 + h(1-\delta)$, and $d_0 = 1 - \delta$. For the right branch, use $f(x) = x^2/2$, $x_0 = 1$, and $d_0 = 1 + \delta$. These are the classical scalar examples in [1]. Their ratios attain the two outer expressions, completing the proof of Theorem 1.

2.3 The published root and the optimal step

The parameter λ in (10) is the largest real root of

$$\begin{aligned} 0 = {} & [(D-1)h^2 + 2h]\lambda^3 + [2(D-1)h^2 + 5h - 4]\lambda^2 \\ & + [(D-1)h^2 + 4h - 4]\lambda + h - 1. \end{aligned} \tag{24}$$

To check the root choice, note that $\lambda(q)$ increases strictly on $(D, 1)$:

$$\lambda'(q) = \frac{(q-D)^2 + D(1-D)}{q^2(1-q)^2} > 0.$$

If a larger real root $\lambda' > \lambda > 0$ existed, set $q' = [2\lambda' + 1 - h(\lambda' + 1)]/[h\lambda'(\lambda' + 1)]$. Its numerator is positive, because $h < 2$ and it is already positive at λ. Equation (24) implies $D = q'[1 - \lambda'(1 - q')]$, so q' is the unique positive root of that quadratic and lies in $(D, 1)$. Hence $q' > q$. But $h(q)$ is strictly decreasing throughout $[q_R, 1)$: above δ, the numerator in (16) is $V^2 + (1-D)(q^2 - D) > 0$. Thus $h(q') < h(q)$, contradicting the defining relation. This proves the root assertion.

Finally, (12) gives

$$\frac{1}{\rho + \frac{1}{2}} = \frac{2\lambda}{h\lambda^2 + (2h-1)\lambda + h - 1},$$

which is exactly the intermediate expression in [1, Theorem 2.3].

Corollary 2.1. *There is a unique $q_* \in (q_R, \delta)$ such that $\delta^2 = q_*^2(2 - q_*)$. The unique minimizer of $\mathcal{W}_1(h, \delta)$ on $[0, 2/(1+\delta)]$ is*

$$h_* = \frac{3 - 2q_*}{(2 - q_*)(1 + q_* - q_*^2)}, \qquad \mathcal{W}_1(h_*, \delta) = \frac{2(1 + q_* - q_*^2)}{(2 - q_*)^2}. \tag{25}$$

For the attaining instance (21)–(22) at this parameter,

$$\langle e_0, d_0 \rangle = 0, \qquad \langle e_0, g_0 \rangle = -\|e_0\|^2 = -\delta^2 \|g_0\|^2 < 0. \tag{26}$$

Proof. The left expression in (5) decreases with h and the right expression increases. In the intermediate interval,

$$\rho'(q) = \frac{D - q^2(2-q)}{q^2(1-D)}.$$

The function $q^2(2-q)$ increases strictly on $(0, 1)$. Its root q_* for the displayed value D is below δ. It lies above q_R, since

$$q_*^2 - (2+\delta)q_* + D + \delta = (1 - q_*)[\delta - q_*(2 - q_*)] < 0.$$

Thus ρ has its unique maximum at q_*. Substitution proves (25). For the extremizer, direct simplification yields

$$\langle e_0, d_0 \rangle = \frac{(1-q)U}{1-D}[D - q^2(2-q)],$$

which vanishes at q_*. The error constraint is attained, and $g_0 \neq 0$, so (26) follows from $g_0 = d_0 - e_0$. ∎

The perpendicularity is to the computed direction. This distinction matters when interpreting the geometric observation in [1, Section 2.4]; it is not perpendicularity to the true gradient. We do not classify every possible extremizing function.

3 The exact asymptotic minimum-gradient curve

We now return to the minimum over all N updated points in (2). Define

$$h_{\max} = \frac{2}{1+\delta}, \qquad h_t = \frac{2}{1+\sqrt{\delta(2-\delta)}}. \tag{27}$$

Since $\delta < \sqrt{\delta(2-\delta)} < 1$, one has $1 < h_t < h_{\max}$. For $0 < h < h_{\max}$, set

$$\Gamma(h,\delta) = \begin{cases} h(1-\delta), & 0 < h \le h_t, \\ \dfrac{h(c^2-\delta^2)}{2u}, \quad c = \dfrac{2}{h} - 1, \quad u = 1 - \sqrt{1-c^2}, & h_t \le h < h_{\max}. \end{cases} \tag{28}$$

The second branch only uses $\delta < c \le \sqrt{\delta(2-\delta)} < 1$. Its parameters satisfy $c^2 = u(2-u)$ and $0 < u \le \delta$. At $h = h_t$, $u = \delta$, so both formulas agree. The function Γ is positive and continuous on the stable interval, and extends continuously by zero at its two endpoints.

Theorem 2. *For every fixed $0 < \delta < 1$ and $0 < h < h_{\max}$,*

$$\lim_{N\to\infty} N\mathcal{W}_N(h,\delta) = \frac{1}{\Gamma(h,\delta)}. \tag{29}$$

For every $N \ge 1$, the lower bound

$$\mathcal{W}_N(h,\delta) \ge \frac{1}{N\Gamma(h,\delta) + \frac{1}{2}} \tag{30}$$

is realized by an explicit scalar or two-dimensional example. The upper limit in (29) *is uniform over all admissible instances at each N.*

The exactness is asymptotic. In particular, (30) is not asserted to be the full finite-horizon curve. We first construct the lower examples and then give a finite-horizon upper estimate with the same leading coefficient.

3.1 Affine travel and transverse oscillation

In the first branch, use $f = \phi_1$, $x_0 = 1 + Nh(1-\delta)$, and $d_k = (1-\delta)g_k$. The actual trajectory is $x_k = 1 + (N-k)h(1-\delta)$ for $0 \le k \le N$. Every gradient is one, and $f(x_0) = Nh(1-\delta) + 1/2$. This gives (30).

For the second branch define

$$t = \frac{\delta^2 - u^2}{c^2 - \delta^2} \ge 0, \qquad b = 1 - u > 0, \tag{31}$$

and take the globally 1-smooth convex function $f(v,w) = \phi_1(v) + w^2/2$. Starting at $x_0 = (1 + Nhb, \sqrt{t})$, choose

$$d_k = \left(b, \frac{2}{h}(-1)^k\sqrt{t}\right). \tag{32}$$

Exact substitution in the recurrence gives

$$x_k = (1 + (N-k)hb, (-1)^k\sqrt{t}), \qquad 0 \le k \le N. \tag{33}$$

The first coordinate remains at least one. Consequently

$$g_k = (1, (-1)^k\sqrt{t}), \qquad e_k = (-u, c(-1)^k\sqrt{t}),$$

and

$$\|e_k\|^2 = u^2 + c^2 t = \delta^2(1+t) = \delta^2\|g_k\|^2.$$

Every true gradient has the same norm. Moreover

$$f(x_0) = Nhb + \frac{1+t}{2}, \qquad \frac{hb}{1+t} = \frac{h(c^2-\delta^2)}{2u} = \Gamma.$$

This proves (30) in the second branch as well. At the interface $t = 0$, so the construction becomes the scalar one. These examples also give terminal-gradient lower bounds; the upper bound in Theorem 2 concerns only the minimum gradient.

3.2 A local certificate with a telescoping correction

The residuals (6)–(8) allow a more flexible local identity than the uncorrected one-step bound. For suitable parameters, we have

$$\begin{aligned} &f(x) - f(x_+) - \Gamma\|g_+\|^2 - \eta(\|g\|^2 - \|g_+\|^2) \\ &\quad = (\lambda+1)R_{01} + \lambda R_{10} + \beta I \\ &\qquad + \beta\|e + \sigma((\lambda+1)g_+ - \lambda g)\|^2 + \zeta\|g - g_+\|^2. \end{aligned} \tag{34}$$

The right-hand multipliers are nonnegative. The coefficient η may have either sign; its two boundary terms will be retained.

For $0 < h \le h_t$, take

$$\begin{aligned} &\sigma = \delta, \qquad \lambda = \frac{2 - h(1+\delta)}{2h\delta}, \qquad \beta = \frac{h}{2\delta}, \\ &\eta = \frac{h}{2}(\delta - 1)(\lambda - 1), \qquad \zeta = \frac{h^2(1-\delta)^2 - 4h + 4}{8\delta h}, \qquad \Gamma = h(1-\delta). \end{aligned} \tag{35}$$

Here $\lambda > 0$. The smaller positive root of the numerator of ζ is h_t, hence $\zeta \ge 0$. Expanding (34) verifies the identity.

For $h_t < h < h_{\max}$, use c, u from (28) and put

$$p = \frac{c}{1 + \sqrt{1 - c^2}} \in (0, 1). \tag{36}$$

Then

$$c = \frac{2p}{1+p^2}, \quad u = \frac{2p^2}{1+p^2}, \quad h = \frac{2(1+p^2)}{(1+p)^2}.$$

The parameters in (34) are

$$\sigma = u, \qquad \lambda = \frac{1-p}{2p}, \qquad \beta = \frac{h}{2u}, \qquad \zeta = 0, \tag{37}$$
$$\eta = \frac{p^4 + p^3 + 3p^2 - p - \delta^2(1+p^2)^2}{2p^2(1+p)^2},$$
$$\Gamma = \frac{4p^2 - \delta^2(1+p^2)^2}{2p^2(1+p)^2} = \frac{h(c^2-\delta^2)}{2u}.$$

All right-hand multipliers are positive except $\zeta = 0$, and $\Gamma > 0$ since $c > \delta$. The rational substitutions above verify (34) directly.

3.3 Uniform control of both boundary terms

It remains to justify a sharp upper bound for the supremum in (2); a pointwise limit along one trajectory would not suffice. For the fixed stable h, δ, set

$$\alpha = \frac{h(1-\delta)^2}{1+\delta}\left(1 - \frac{h(1+\delta)}{2}\right) > 0. \tag{38}$$

The error condition gives

$$(1+\delta)\langle g, d\rangle - \|d\|^2 = \delta\|g\|^2 - (1-\delta)\langle g, e\rangle - \|e\|^2 \geq 0.$$

Smoothness and $\|d\| \geq (1-\delta)\|g\|$ therefore imply

$$f(x_k) - f(x_{k+1}) \geq \alpha\|g_k\|^2, \qquad \sum_{k=0}^{N-1}\|g_k\|^2 \leq \frac{f(x_0)}{\alpha}. \tag{39}$$

Choose integers $m \geq 1$ and $N > 2m$. From (39), there exist

$$j \in \{1, \dots, m\}, \qquad \ell \in \{N-m, \dots, N-1\}$$

such that $\|g_j\|^2, \|g_\ell\|^2 \leq f(x_0)/(\alpha m)$. Sum (34) from j to $\ell - 1$ and discard its nonnegative right side. Since f decreases and $f \geq 0$,

$$\begin{aligned}\Gamma \sum_{k=j+1}^{\ell}\|g_k\|^2 &\leq f(x_j) - f(x_\ell) - \eta(\|g_j\|^2 - \|g_\ell\|^2) \\ &\leq f(x_0)\left(1 + \frac{2|\eta|}{\alpha m}\right).\end{aligned} \tag{40}$$

The sum contains at least $N - 2m$ gradients from the index set in (2). Thus, uniformly over every admissible instance,

$$\mathcal{W}_N(h, \delta) \leq \frac{1 + 2|\eta|/(\alpha m)}{\Gamma(h, \delta)(N - 2m)}. \tag{41}$$

Taking $m = \lfloor\sqrt{N}\rfloor$ for sufficiently large N proves the upper limit in (29). The lower examples prove the matching lower limit, completing Theorem 2.

4 Asymptotically optimal constant steps

The next result allows the selected constant step to depend on the horizon. Each individual run still uses one unchanged step.

Theorem 3. *Fix $0 < \delta < 1$. There is a unique $p_* \in (0,1)$ satisfying*

$$\delta^2 = \frac{4p_*^3}{(1+p_*^2)(1+2p_*-p_*^2)}. \tag{42}$$

Define

$$h_\infty = \frac{2(1+p_*^2)}{(1+p_*)^2}, \qquad C_\delta = \frac{1+2p_*-p_*^2}{2(1-p_*)}. \tag{43}$$

Then $h_t < h_\infty < h_{\max}$, and

$$\lim_{N\to\infty} N \inf_{h\ge 0} \mathcal{W}_N(h,\delta) = C_\delta. \tag{44}$$

If $h_N \ge 0$ satisfies $N\mathcal{W}_N(h_N,\delta) \to C_\delta$, then $h_N \to h_\infty$. This includes exact minimizing steps whenever such minimizers exist.

Proof. Let $B(p) = (1+p^2)(1+2p-p^2)$. The right side of (42) increases from zero to one on $(0,1)$, because

$$\frac{d}{dp}\frac{4p^3}{B(p)} = \frac{4p^2(p+1)^2(p^2-2p+3)}{B(p)^2} > 0.$$

This proves existence and uniqueness. At this root, the substitutions (36) give

$$\frac{\delta^2}{c^2} = \frac{p(1+p^2)}{1+2p-p^2} < 1, \qquad \frac{\delta^2}{u^2} = \frac{1+p^2}{p(1+2p-p^2)} > 1.$$

The positive differences in the two comparisons are $(1-p)(1+p)^2$ and $(1-p)^2(1+p)$, respectively. Thus $u < \delta < c$, which places h_∞ in the open second branch.

In the p parametrization of that branch,

$$\frac{d\Gamma}{dp} = \frac{\delta^2 B(p) - 4p^3}{p^3(1+p)^3}. \tag{45}$$

The sign changes once, from positive to negative at p_*. Since h decreases strictly with p, Γ has its unique maximum there. Its first branch increases with h, so this maximum is global. Substitution yields

$$\Gamma_{\max} = \Gamma(h_\infty,\delta) = \frac{2(1-p_*)}{1+2p_*-p_*^2} = \frac{1}{C_\delta}. \tag{46}$$

For stable h, the actual lower instances in (30) give $\mathcal{W}_N(h,\delta) \ge [N\Gamma_{\max} + 1/2]^{-1}$. At $h = 0$, a scalar quadratic gives ratio two. For $h \ge h_{\max}$, the same quadratic with

$d_k = (1+\delta)g_k$ has minimum-gradient ratio at least two, because the magnitude of its iteration factor is at least one. Hence the lower bound

$$\inf_{h\ge 0} \mathcal{W}_N(h,\delta) \ge \frac{1}{N\Gamma_{\max} + \frac{1}{2}}$$

holds on the whole noncompact step range. The fixed choice $h = h_\infty$ in Theorem 2 proves the matching upper limit, establishing (44).

Finally, an asymptotically minimizing sequence must be stable and positive for all sufficiently large N, since otherwise $N\mathcal{W}_N \ge 2N$ along a subsequence. The lower bound (30) forces $\Gamma(h_N,\delta) \to \Gamma_{\max}$. Extend Γ by zero at both endpoints of $[0, h_{\max}]$. Its continuity, the compactness of that interval, and the unique maximum proved above imply $h_N \to h_\infty$. ∎

The two optimal steps solve different questions. The step h_* in Corollary 2.1 is exactly optimal for one update. The step h_∞ in Theorem 3 is optimal in the asymptotic minimum-gradient problem. No convergence of the finite-horizon terminal-gradient optimal steps is implied. The finite-horizon examples start from points depending on N; they do not assert a nonconvergent infinite run from one fixed initialization.

5 A dimension-independent last-gradient bound fails

Define the terminal-gradient performance

$$\mathcal{W}_N^{\text{last}}(h,\delta) = \sup \frac{\|\nabla f(x_N)\|^2}{L(f(x_0) - f_*)}, \tag{47}$$

with the same class of instances as in (2). The supremum includes all finite dimensions. By Theorem 2, the minimum-gradient criterion at every $0 < h \le 1$ satisfies $N\mathcal{W}_N(h,\delta) \to [h(1-\delta)]^{-1}$. The corresponding last-gradient quantity behaves differently.

Theorem 4. *Fix $0 < \delta < 1$ and $0 < h \le 1$, and put $r = 1 + h^2\delta^2/4 > 1$. For every integer $m \ge 1$, let $N = 2^{m-1}$. There is a globally convex, continuously differentiable, 1-smooth coercive function on $\mathbb{R}^{m+1}$, with unique minimizer zero, and an admissible run of (1) with $L = 1$ and normalized step h, such that*

$$f(x_0) - f_* = hN + \tfrac{1}{2}, \qquad \|\nabla f(x_N)\|^2 = r^m. \tag{48}$$

Consequently

$$\mathcal{W}_N^{\text{last}}(h,\delta) \ge \frac{r^m}{hN + \frac{1}{2}} = \frac{rN^{\log_2 r}}{hN + \frac{1}{2}}, \qquad N = 2^{m-1}, \tag{49}$$

and

$$\limsup_{N\to\infty} N\mathcal{W}_N^{\text{last}}(h,\delta) = +\infty. \tag{50}$$

Thus no finite constant $C(h,\delta)$ can bound $\mathcal{W}_N^{\text{last}}(h,\delta)$ by $C(h,\delta)/N$ uniformly over this class, even for coercive objectives with unique minimizers.

Proof. Use coordinates $0, \dots, m$ and define positive amplitudes

$$a_0 = 1, \qquad a_j = \frac{h\delta}{2} r^{(j-1)/2} \quad (1 \le j \le m), \qquad f_m(x) = \sum_{j=0}^{m} \phi_{a_j}(x_j), \tag{51}$$

where ϕ_a is (20). The gradient is coordinatewise clipping, so f_m is globally convex, C^1 and 1-smooth. All amplitudes are positive. Therefore its unique minimizer is zero, and

$$f_m(x) \ge \left(\min_j a_j \right) \|x\|_1 - \tfrac{1}{2} \sum_{j=0}^{m} a_j^2$$

proves coercivity.

Split the N updates into m consecutive blocks, with lengths

$$n_j = 2^{m-j-1} \quad (1 \le j < m), \qquad n_m = 1. \tag{52}$$

For $m = 1$ there is only the last formula. Put $t_0 = 0$ and $t_j = \sum_{i=1}^{j} n_i$. Then

$$t_m = N, \qquad N - t_j = n_j \quad (j < m). \tag{53}$$

Initialize $x_0 = (hN + 1, 0, \dots, 0)$. If E_j denotes the coordinate unit vector with index j, prescribe

$$e_k = \frac{2a_j}{h} E_j, \qquad d_k = \nabla f_m(x_k) + e_k, \qquad t_{j-1} \le k < t_j. \tag{54}$$

These errors are specified before the run.

Coordinate zero equals $h(N - k) + 1$ throughout the run, staying at least one. Coordinate j is zero before its block. Its first active update sends it to $-2a_j$. Each subsequent active update subtracts $(2 - h)a_j$, so

$$x_{t_{j-1}+s,j} = -[(2 - h)s + h]a_j, \qquad 1 \le s \le n_j. \tag{55}$$

After that block the coordinate receives no error and moves right by ha_j per update as long as it stays at most $-a_j$. For $j < m$, the resulting formula is

$$x_{k,j} = -[(2 - h)n_j + h - h(k - t_j)]a_j, \qquad t_j \le k \le N. \tag{56}$$

Its largest value occurs at N. By (53),

$$x_{N,j} = -[2(1 - h)n_j + h]a_j \le -a_j,$$

since $n_j \ge 1$ and $h \le 1$. Thus the entire post-block formula is consistent with the clipped gradient. These formulas prove the actual trajectory by induction, including every junction contact. The last coordinate equals $-2a_m$ at time N, and hence

$$\nabla f_m(x_N) = (1, -a_1, \dots, -a_m).$$

At the start of block j, the preceding coordinates have gradients $1, -a_1, \dots, -a_{j-1}$, while coordinate j and all later coordinates have zero gradient. Their squared norm is

$$1 + \sum_{i=1}^{j-1} a_i^2 = r^{j-1}. \tag{57}$$

The error in (54) has norm $2a_j/h = \delta r^{(j-1)/2}$, so it satisfies the relative-error condition with equality at block entry. After the first update in that block, the squared gradient norm is r^j, while the error norm is unchanged. The condition is then strict. This verifies every prescribed direction.

Only coordinate zero contributes to the initial value: $f_m(x_0) = \phi_1(hN+1) = hN+1/2$. The final squared gradient is $1 + \sum_{j=1}^{m} a_j^2 = r^m$. This proves (48) and (49). Since $r > 1$, multiplying (49) by N and letting $m \to \infty$ proves (50). ∎

Every step in this theorem is strictly below $2/(1+\delta)$. The dimension is $m + 1 = 2 + \log_2 N$, so Theorem 4 does not exclude a bound with a dimension-dependent constant. It does not assert a slow infinite trajectory of one fixed instance. At $h = 1$, setting all subsequent errors to zero brings the displayed endpoint to its minimizer in two more updates: coordinate zero and all previously completed coordinates become zero after one update, while the last coordinate moves from $-2a_m$ to $-a_m$ and then to zero. For every h in the theorem, the minimum squared gradient among $x_1, \dots, x_N$ is exactly r, so the normalized minimum-gradient ratio is $r/(hN + 1/2)$.

Gradient amplification itself is an established antecedent: [4, Theorem 2.8 and the discussion on pp. 39–41] studies it and obtains a non-tight last-gradient estimate by summing geometric weights. The obstruction above uses complete actual trajectories whose horizons and dimensions grow together. Its lower exponent is not claimed to be the sharp replacement for the failed $1/N$ rate.

6 Planar last-gradient estimates

Restricting the dimension to two changes the conclusion at smaller steps. The following bound uses the initial function gap and is uniform over all objectives, starts and admissible errors in the stated planar class.

Theorem 5. *Let $f : \mathbb{R}^2 \to \mathbb{R}$ satisfy the assumptions of Section 1. Fix $0 < \delta < 1$ and*

$$0 < h \le \frac{1}{1+\delta}.$$

Set

$$K_\delta = \frac{2\sqrt{\delta}}{1-\delta}, \qquad B_\delta = e^{2\pi K_\delta}, \qquad \alpha_h = h(1-\delta)\left(1 - \frac{h(1-\delta)}{2}\right). \tag{58}$$

Every run of (1) *satisfies*

$$\|\nabla f(x_j)\| \le B_\delta \|\nabla f(x_i)\| \qquad (0 \le i \le j). \tag{59}$$

For every $N \geq 1$ and $f(x_0) > f_*$,

$$\frac{\|\nabla f(x_N)\|^2}{L(f(x_0) - f_*)} \leq \frac{1}{\frac{1}{2} + \alpha_h N / B_\delta^2}. \tag{60}$$

No boundedness assumption on the sublevels or trajectory is required.

The constant is not claimed to be sharp. Theorem 4 excludes a dimension-independent bound at every step in Theorem 5, so the two statements compare the same algorithm and step. They do not determine what happens in each fixed dimension greater than two.

6.1 A bound from newly visited gradient rays

We first isolate the planar argument. A continuous nonzero planar curve on a compact interval has a continuous real-valued angle lift. For an absolutely continuous curve, an absolutely continuous lift is obtained by integrating $\det(g, g')/\|g\|^2$ and fixing the initial angle.

Lemma 1. *Let $g : [0, T] \to \mathbb{R}^2 \setminus \{0\}$ be absolutely continuous, and write*

$$g(t) = e^{u(t)}(\cos\theta(t), \sin\theta(t)).$$

Suppose a later visit to the same oriented ray cannot increase the norm:

$$\theta(t) - \theta(s) \in 2\pi\mathbb{Z}, \quad s < t \quad \Longrightarrow \quad u(t) \leq u(s). \tag{61}$$

Let $U(t) = \max_{0 \leq s \leq t} u(s)$. Assume that, almost everywhere on the set $\{u = U,\ u' > 0\}$,

$$u'(t) \leq K|\theta'(t)| \tag{62}$$

for a fixed $K \geq 0$. Then

$$\max_{0 \leq t \leq T} \|g(t)\| \leq e^{2\pi K} \|g(0)\|.$$

Proof. Let $a(t) = \min_{s \leq t} \theta(s)$, $b(t) = \max_{s \leq t} \theta(s)$ and $w(t) = b(t) - a(t)$. The running extrema of an absolutely continuous scalar function are absolutely continuous: their variation on an interval is bounded by the integral of the absolute derivative there. Thus U, w are absolutely continuous and nondecreasing.

Work at a time where all these functions are differentiable. If $U' > 0$, then $u = U$ and $u' = U'$: a strict gap below the running maximum would make U locally constant, and the derivative of $U - u$ at its zero is zero. If the current ray had been visited at some $s < t$, (61) would force equality throughout

$$u(t) \leq u(s) \leq U(s) \leq U(t) = u(t).$$

Then U would be constant on $[s, t]$, contradicting $U'(t) > 0$. Therefore a growing record must visit a new ray.

Continuity shows that an angle strictly between $a(t)$ and $b(t)$ was already visited. Hence growing records occur only at an extreme of the lifted angle. At $\theta = b$, differentiation of $b - \theta$ at its zero gives $\theta' = b' \geq 0$; at $\theta = a$, it gives $\theta' = a' \leq 0$. In either case $|\theta'| \leq w'$. Once $w \geq 2\pi$, every ray has already been visited: at an angular extreme use the earlier angle differing by 2π, and in the interior use continuity. Thus, almost everywhere,

$$0 \leq U' \leq Kw'\mathbf{1}_{\{w<2\pi\}}. \tag{63}$$

Equivalently, $U - K\min\{w, 2\pi\}$ is nonincreasing. Integration and $w(0) = 0$ give $U(T) - U(0) \leq 2\pi K$, which proves the result. ∎

For gradients of one smooth convex function, the ray condition follows from objective descent. Indeed, if $f(y) \leq f(x)$ and $\nabla f(y) = \lambda \nabla f(x) \neq 0$ with $\lambda > 1$, cocoercivity gives

$$\langle \nabla f(x), y - x \rangle \geq \frac{\lambda - 1}{L} \|\nabla f(x)\|^2 > 0.$$

Convexity would imply $f(y) > f(x)$, a contradiction. Consequently any path along which f is nonincreasing satisfies (61).

6.2 The actual polygonal path and the endpoint

Proof of Theorem 5. Normalize $L = 1$. Write $g_k = \nabla f(x_k)$ and $d_k = g_k + e_k$. On one actual step, put

$$G(q) = \nabla f(x_k - qd_k), \qquad 0 \leq q \leq h.$$

The error inequality and cocoercivity respectively give

$$\left\| g_k - \frac{d_k}{1 - \delta^2} \right\| \leq \frac{\delta}{1 - \delta^2} \|d_k\|, \tag{64}$$

$$\left\| G(q) - g_k + \frac{q}{2} d_k \right\| \leq \frac{q}{2} \|d_k\|. \tag{65}$$

For (65), expand $\|G(q) - g_k\|^2 \leq -q\langle G(q) - g_k, d_k \rangle$. Adding the two balls places $G(q)$ in the ball with center $[1/(1 - \delta^2) - q/2]d_k$ and radius $[\delta/(1 - \delta^2) + q/2]\|d_k\|$. These balls are nested as q increases. At $q = 1/(1 + \delta)$ the center coefficient and radius coefficient both equal $1/[2(1 - \delta)]$. Hence, including at the step endpoint,

$$\left\| G(q) - \frac{d_k}{2(1 - \delta)} \right\| \leq \frac{\|d_k\|}{2(1 - \delta)}, \qquad \langle G(q), d_k \rangle \geq (1 - \delta)\|G(q)\|^2. \tag{66}$$

The second inequality is just the squared first inequality. In particular, f is nonincreasing along the entire spatial segment.

Join all the actual steps by their affine interpolations and restrict this path to any finite interval of vertices $i, \ldots, j$. If the final gradient is zero, (59) is immediate. Otherwise the gradient is nonzero everywhere on this path: a zero gradient would attain f_*, after which

the objective could not decrease further and every later gradient would be zero. A zero-gradient vertex is stationary because its error is zero.

The gradient path is absolutely continuous. At almost every time, monotonicity of ∇f, applied to difference quotients, gives

$$\langle g'(t), x'(t)\rangle \geq 0.$$

On the interior of an edge this means $\langle g', d_k\rangle \leq 0$; no Hessian is assumed to exist. Write $g = rn$, $n = (\cos\theta, \sin\theta)$, and let Jn be the counterclockwise rotation of n by $\pi/2$.

At a running norm record on this restricted path, $r \geq \|g_k\|$ because the current edge starts at an earlier vertex. Since $\|d_k\| \leq (1+\delta)\|g_k\|$, (66) gives

$$\frac{\langle g, d_k\rangle}{r\|d_k\|} \geq \frac{1-\delta}{1+\delta}. \tag{67}$$

Resolve $d_k = An + CJn$. Then $A > 0$ and $|C|/A \leq 2\sqrt{\delta}/(1-\delta) = K_\delta$. The inequality $\langle g', d_k\rangle \leq 0$ therefore yields, at these record times,

$$(\log r)' = \frac{r'}{r} \leq -\frac{C}{A}\theta' \leq K_\delta|\theta'|.$$

The objective is nonincreasing on the full path, so the ray condition holds. Lemma 1, restarted at vertex i, proves (59). The argument requires a uniform angle only at norm records; it remains valid when the all-points angle degenerates at $h = 1/(1+\delta)$.

It remains to sum descent. Since $h \leq 1/(1+\delta) < 1$, smoothness gives

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq \left(h - \frac{h^2}{2}\right)\|g_k\|^2 + h(1-h)\langle g_k, e_k\rangle - \frac{h^2}{2}\|e_k\|^2 \\ &\geq \alpha_h\|g_k\|^2. \end{aligned} \tag{68}$$

Use $\|g_k\|^2 \geq B_\delta^{-2}\|g_N\|^2$ for $0 \leq k < N$ and the terminal bound $f(x_N) - f_* \geq \|g_N\|^2/2$. Their sum is

$$f(x_0) - f_* \geq \left(\frac{1}{2} + \frac{\alpha_h N}{B_\delta^2}\right)\|g_N\|^2.$$

Restoring L proves (60). ∎

6.3 Continuous relative-error flow

The same planar mechanism has a direct continuous version with a smaller constant.

Proposition 6.1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ satisfy the assumptions of Section 1, and let x be a locally absolutely continuous solution of

$$x'(t) = -\nabla f(x(t)) - e(t), \qquad \|e(t)\| \leq \delta\|\nabla f(x(t))\| \quad \text{almost everywhere,}$$

with measurable e. Put $\kappa_\delta = \delta/\sqrt{1-\delta^2}$. Then, for all $0 \leq s \leq t$,

$$\|\nabla f(x(t))\| \leq e^{2\pi\kappa_\delta}\|\nabla f(x(s))\|. \tag{69}$$

For every $T > 0$,

$$\|\nabla f(x(T))\|^2 \le \frac{e^{4\pi\kappa_\delta}}{(1-\delta)T}\big(f(x(0)) - f_*\big). \tag{70}$$

Proof. Write $g = \nabla f(x)$. On a compact interval where $g \neq 0$, the chain rule gives $f' = -\|g\|^2 - \langle g, e\rangle \le -(1-\delta)\|g\|^2$. Thus the ray condition holds. Monotonicity, differentiated only along the absolutely continuous paths, gives

$$\langle g', g\rangle \le -\langle g', e\rangle \le \delta\|g'\|\|g\|.$$

Writing $g = r(\cos\theta, \sin\theta)$, whenever $r' > 0$ this implies

$$r' \le \delta\sqrt{(r')^2 + r^2(\theta')^2}, \qquad (\log r)' \le \kappa_\delta|\theta'|.$$

Lemma 1 proves (69) on every such interval. If a zero gradient is reached at x_*, the estimate $\|x'\| \le (1+\delta)L\|x - x_*\|$ and Gronwall's inequality show that the solution remains there. This also covers zero initial gradients.

Finally, apply (69) from each $s \in [0, T]$ to T, and integrate the descent inequality:

$$f(x(0)) - f_* \ge (1-\delta)\int_0^T \|g(s)\|^2\, ds \ge (1-\delta)Te^{-4\pi\kappa_\delta}\|g(T)\|^2.$$

This proves (70). ∎

The discrete planar result includes the smaller-step endpoint, but it does not cover $h = 1$ for $\delta > 0$. The constant-step planar last-gradient rate at that step, the largest admissible planar step range, and sharp constants remain undetermined here. The continuous statement and spatial length results do not by themselves settle these discrete questions.

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